

TEST OF MATHEMATICS FOR UNIVERSITY ADMISSION

PAPER 1: APPLICATIONS OF MATHEMATICAL KNOWLEDGE

Practice paper · Set 05

75 minutes

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but do not open the question paper until you are told that you may do so.

A separate answer sheet is provided for this paper. You require a soft pencil and an eraser. Please complete the answer sheet with your candidate number, centre number, date of birth and full name.

This paper contains **20 questions**. For each question, choose the **one** answer you consider correct and record your choice on the separate answer sheet. If you make a mistake, erase thoroughly and try again.

There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all 20 questions. Each question is worth one mark.

You can use the question paper for rough working or notes, but no extra paper is allowed.

You must complete the answer sheet within the time limit.

Calculators and dictionaries are NOT permitted.

There is no formulae booklet for this test.

Please wait to be told you may begin before turning this page.

This is a practice paper generated from a question bank written to the TMUA content specification. It is not an official UAT-UK paper. Answers and worked solutions are in the accompanying solutions booklet.

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1 The curve $y = x^3 + px^2 + qx$ has stationary points at $x = 1$ and $x = 3$. Find q .

A 3

B 12

C 9

D -9

E -6

F 6

2 The expression $x^2 + 6x + c$ can be written as $(x + p)^2 + q$. Given that $q = -c$, find the value of c .

A 9

B 3

C $-\frac{9}{2}$

D $\frac{9}{2}$

E 0

F -9

- 3** A mixture contains sand and cement in the ratio 5 : 2. How much cement must be added to 35 kg of the mixture to make the ratio 5 : 3?
- A** 2.5 kg
 - B** 15 kg
 - C** 3 kg
 - D** 7 kg
 - E** 5 kg
 - F** 10 kg

- 4 Evaluate $\frac{(3 \times 10^4) \times (8 \times 10^{-2})}{6 \times 10^{-3}}$, giving your answer in standard form.
- A** 4×10^4
- B** 4×10^6
- C** 4×10^{-1}
- D** 2.4×10^5
- E** 1.5×10^5
- F** 4×10^5

5 The graph of $y = \log_2 x$ is reflected in the line $y = x$. The resulting graph has equation

A $y = 2^x$

B $y = \log_{\frac{1}{2}} x$

C $y = \log_2 (-x)$

D $y = 2^{-x}$

E $y = x^2$

F $y = -\log_2 x$

6 The points $A(1, 2)$, $B(5, 4)$ and $C(k, 8)$ are collinear. Find k .

A 12

B 11

C 10

D 9

E 7

F 13

7 The polynomial $p(x) = x^3 + ax^2 + bx + 4$ has $(x - 1)$ and $(x + 2)$ as factors. Find the third linear factor.

A $(x + 1)$

B $(x - 2)$

C $(x - 4)$

D $(x - 1)$

E $(x + 4)$

F $(x + 2)$

- 8** A bag contains 4 red and 6 blue counters. Two counters are taken at random without replacement. Find the probability that they are different colours.

A $\frac{24}{100}$

B $\frac{15}{2}$

C $\frac{5}{8}$

D $\frac{15}{4}$

E $\frac{12}{12}$

F $\frac{25}{25}$

9 Solve $xy = 6$ and $x - y = 1$. What is the sum of all possible values of $x + y$?

A 10

B 5

C -5

D 7

E 1

F 0

10 Find the coefficient of x^2 in the expansion of $(x + 1)(x - 2)(2x + 3)$.

A 1

B -3

C -2

D -1

E 2

F 3

- 11** Two sides of a triangle are 3 and 5 and the included angle is θ , where $\cos \theta = -\frac{1}{5}$. Find the length of the third side.

A $\sqrt{34}$

B 6

C $\sqrt{40}$

D 7

E $\sqrt{28}$

F $\sqrt{31}$

12 By completing the square, find the minimum value of $2x^2 - 12x + 23$.

A -13

B 5

C 23

D 7

E -5

F 3

13 How many times does the graph of $y = x^3 - x^2 - 6x$ cross or touch the x -axis?

A 3

B 1

C 0

D 2

E 4

14 Find the value of $\int_1^8 x^{-\frac{2}{3}} dx$.

A $\frac{21}{2}$

B 9

C 3

D 6

E 2

F $\frac{3}{2}$

- 15** The graph of $y = \sin x$ is transformed to $y = 2\sin(3x) + 1$. What is the maximum value of the new function, and what is its period (in degrees)?
- A** maximum 1, period 120°
 - B** maximum 3, period 360°
 - C** maximum 2, period 120°
 - D** maximum 3, period 120°
 - E** maximum 3, period 1080°
 - F** maximum 2, period 180°

16 A cylinder has radius r and height h . Its volume is 54π and $h = 2r$. Find its total surface area.

- A** 27π
- B** 45π
- C** 108π
- D** 54π
- E** 72π
- F** 36π

17 Solve $2\cos^2 x + 3\sin x - 3 = 0$ for $0^\circ \leq x \leq 360^\circ$. How many solutions are there?

A 2

B 0

C 4

D 5

E 3

F 1

- 18** The lines $y = mx + 4$ and $y = 3x - 2$ intersect at a point with positive x - and y -coordinates. Find the complete set of values of m .
- A** $m < 3$
- B** $m < -6$ or $m > 3$
- C** $-6 < m < 3$
- D** $m > -6$
- E** $0 < m < 3$
- F** $-6 < m < 0$

19 The graph of $y = 2^x$ is translated 3 units in the positive x -direction. The resulting graph has equation

A $y = \frac{1}{8} \cdot 2^x$

B $y = 2^{x/3}$

C $y = 8 \cdot 2^x$

D $y = 2^x - 3$

E $y = 2^{3x}$

F $y = 2^x + 3$

20 The line $y = x + c$ is a tangent to the circle $(x - 1)^2 + (y - 2)^2 = 8$. Find the possible values of c .

A $c = -3$ or $c = 5$

B $c = 1 \pm 2\sqrt{2}$

C $c = 1$ or $c = 5$

D $c = -1$ or $c = 3$

E $c = \pm 4$

F $c = -5$ or $c = 3$

END OF TEST